My Reproducible Manuscript

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This is a manuscript to show that I have learned to:

- Use markup language(s)
- Use figures
- Use tables
- Use equations
- Reference
- Make a document with clear presentation and consistency

Dog Intelligence

I will demonstrate my reproducible markup skills by using dog intelligence as an example, using data from Beautiful (2026).

I can use code that is not shown in the rendered document, by including `echo = FALSE`, like this:

```
See the qmd file for the code that is hidden in the rendered document
```

Or I can choose to use code that is shown in the rendered document, like this:

```r
# Load data
# make sure only relative paths are used to ensure reproducibility!
DogInt <- read_csv("../data/dog_intelligence.csv")
BreedInf <- read_csv("../data/AKC Breed Info.csv")
DogCateg <-  read_excel("../data/DOG_categories.xlsx")

# Join the datasets based on breed
DogData <- left_join(DogInt, BreedInf, by = "Breed")
DogData <- left_join(DogData, DogCateg, by = "Breed")
```

```
# Remove double index variable
DogData <- DogData %>% dplyr::select(-index.y)
```

Table

As a measure of dog intelligence, I will look at the amount of repetitions they need before obeying. The more repetitions are needed, the less intelligent a dog is. Because the dataset contains an enormous amount of dog breeds, with 136 different breeds, I decided to only look at dog categories, which contains 8 different categories.

Some descriptive statistics about the repetitions needed before obeying a command, per category, can be found in the table below:

Table 1: Repetitions of Command Needed Before Obeying Per Dog Category

Category	Mean Lower Reps	Mean Upper Reps	Mean Reps
herding	10.66667	20.73333	15.70000
hound	41.50000	56.50000	49.00000
non-sporting	31.20000	49.26667	40.23333
sporting	12.56522	22.52174	17.54348
terrier	26.18182	46.13636	36.15909
toy	34.26667	56.93333	45.60000
working	24.22222	39.88889	32.05556
NA	25.87500	45.00000	35.43750

Reps = repetitions; lower = least nr. of reps before obeying; upper = highest nr. of reps before obeying

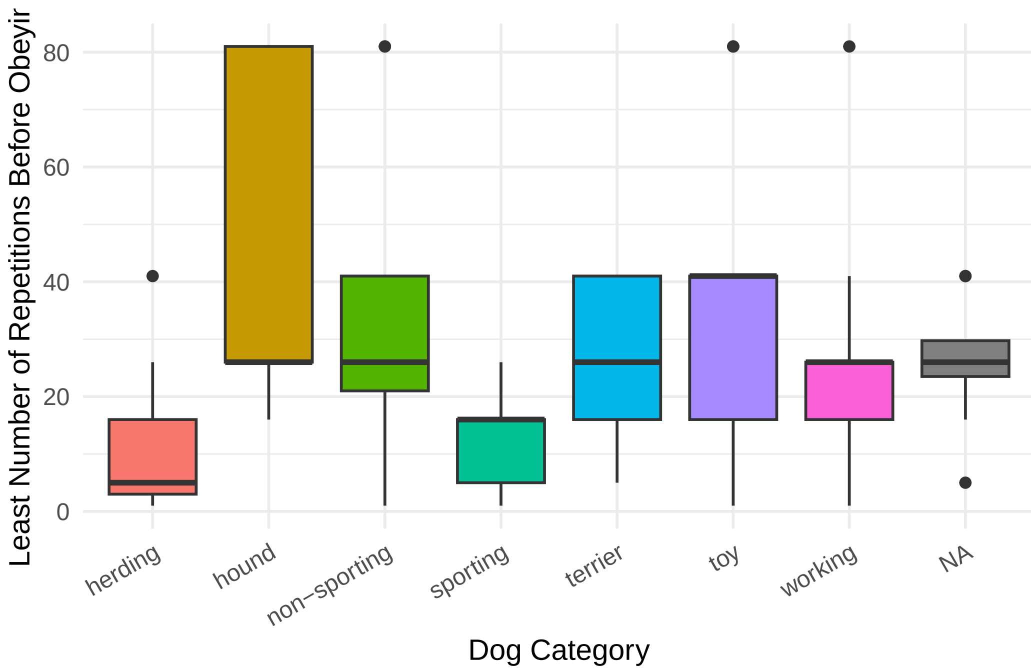
Figure

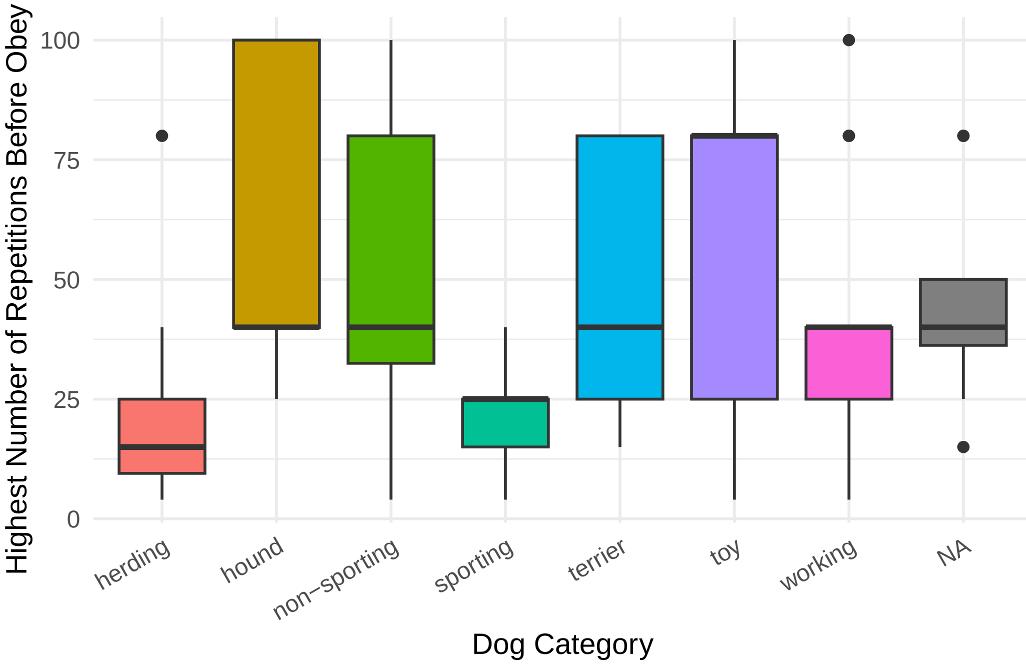
To show I can make a figure, I will plot dog intelligence as a function of dog category.

```
# The lowest number of repetitions before a dog understands the command
ggplot(data = DogData,
       aes(x = Category,
                    y = reps_lower,
                    fill = Category)) +
  geom_boxplot() +
  theme_minimal()  +
  theme(
    # remove legend
```

```
    legend.position = "none",

    # create label that is presented in a slight angle to increase readability
    axis.text.x = element_text(angle = 30, hjust = 1)
  ) +
  labs(x = "Dog Category",
       y = "Least Number of Repetitions Before Obeying")
```





It is clear that herding and sporting dogs need the lowest number of repetitions, while on average hounds need the most repetition.

Equations

Hypothetically, we could say that dog intelligence per category ($I_{category}$) can be calculated as a function of the mean number of repetitions needed before obeying, as shown in Equation 1.

$$I_{\text{category}} = \frac{1}{1 + Rep_{\text{category}}} \tag{1}$$

In Equation 1, Rep_{category} is the mean number of repetitions needed per category.

If we want a more complex (hypothetical!) model for dog intelligence per category ($I_{category}$), we could even include learning speed ($Speed_{category}$) and some constants (α), as shown in Equation 2.

$$I_{\text{category}} = \alpha \cdot \frac{1}{1 + Rep_{\text{category}}} + (1 - \alpha) \cdot Speed_{\text{category}} \tag{2}$$

Sourcing scripts

These equations can be implemented in a separate R script, and ran in this manuscript, by sourcing the R script `intelligence_eq1.R`, giving:

```
# Use relative path to source the script
source("../scripts/intelligence_eq1.R")
```

With this code, we can calculate the intelligence per category (using Eq.1) using the `DogSummary` dataset, `intelligence(DogSummary$mean_total_rep)`, giving: 0.0598802, 0.02, 0.0242522, 0.0539273, 0.0269113, 0.0214592, 0.0302521, 0.0274443.

This shows that herding dogs are indeed most intelligent, based on the number of repetitions they need before obeying a command.

Images of the smartest dogs

Not only the smartest, they might also be the cutest ("Dog - Herding, Breeds, Training | Britannica" 2026):

Figure 1: Herding dog puppies

For a very nice overview of more data on different dog breeds, see this website (Beautiful 2026).

References

Beautiful, Information is. 2026. "Best in Show – What's the Best Dog Breed, According to Data?" *Information Is Beautiful*. https://informationisbeautiful.net/visualizations/best-in-show-whats-the-top-data-dog/.

"Dog - Herding, Breeds, Training | Britannica." 2026. https://www.britannica.com/animal/dog/Herding-dogs.